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Python Programming Foundations

Week 5 - Files, Errors, & Data Persistence
Day 1: "Programs That Remember"

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Programs Can Remember



Until now, most of our programs have only worked while they were running.

Today we add persistence: the program can save information outside itself and load that information again later.

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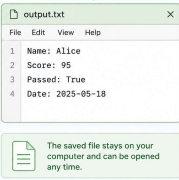
Program Ends, File Remains

program run ends



The program window closes when the run ends.

saved file remains



The saved file stays on your computer and can be opened any time.

Programs Can Remember

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Quick Review

Printing a value shows it, but it does not save it.

If the program ends and the value only lived in a variable, that value is gone.



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The Save / Load Cycle

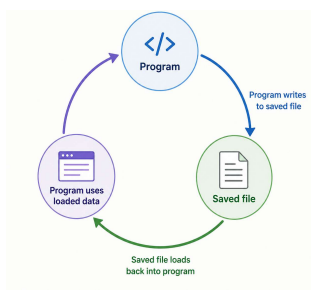


Persistence is a cycle:

- ▶ create or collect information
- ▶ write it to a file
- ▶ close or rerun the program
- ▶ read the file back
- ▶ display or use the loaded information

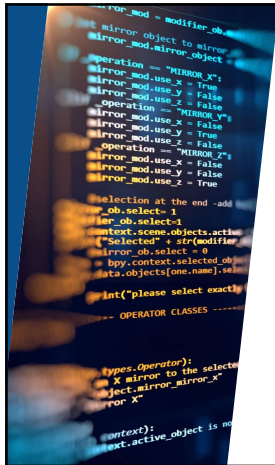
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The Save-Load Cycle

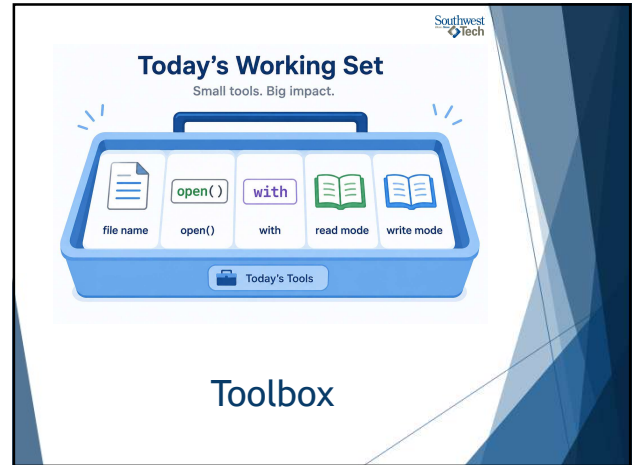
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What We Will Use Today

- ▶ a file name
- ▶ `'open()'`
- ▶ `'with'`
- ▶ write mode
- ▶ read mode
- ▶ visible output after loading

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Today's Working Set

Small tools. Big impact.

file name open() with read mode write mode

Today's Tools

Toolbox

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What We Will "Skip" For Now

We will save these for later:

- ▶ databases
- ▶ binary files
- ▶ file compression
- ▶ custom JSON encoding
- ▶ complex folder and deployment issues

Today's goal is simple: save text, load text, explain the cycle.

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– Parked for Later –

binary files file compression custom JSON encoding complex folder deployment issues

Parked for Later

Useful later, not today's required target.

Parked For Later

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Review Coding Lab

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What Counts as Success Today

 WRITING A FILE
  READING A FILE
  EXPLAINING THE CYCLE

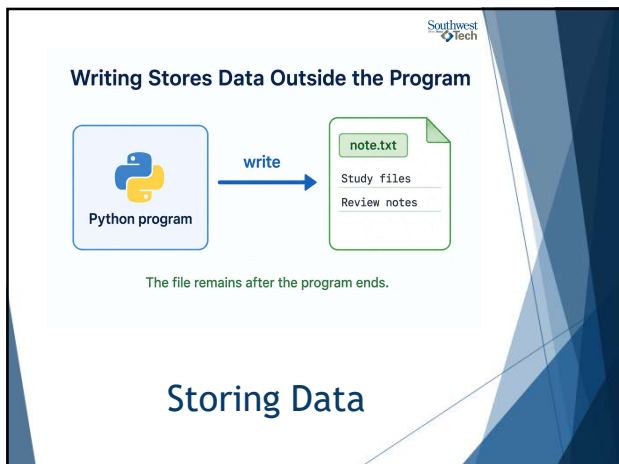
12



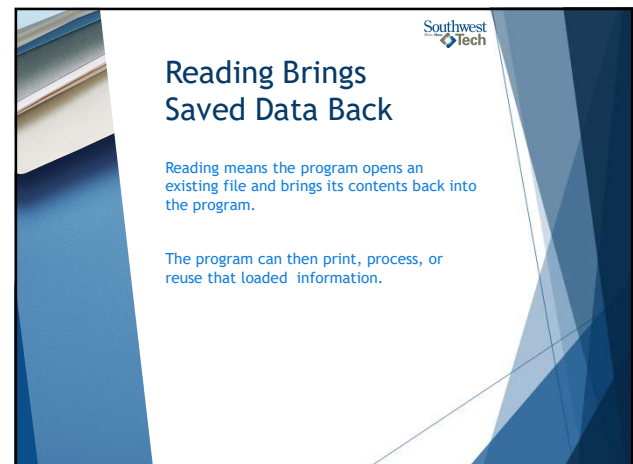
13



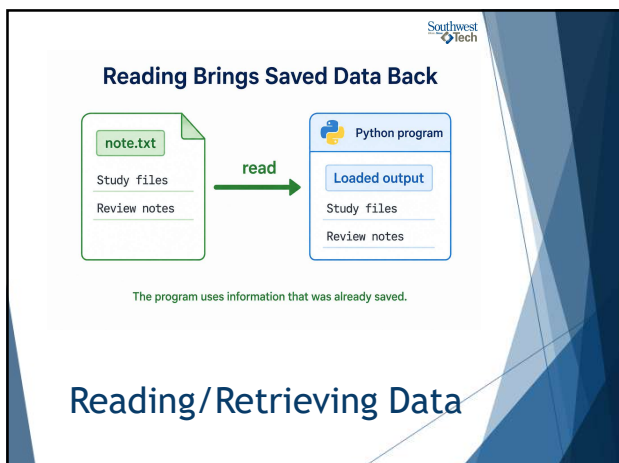
14



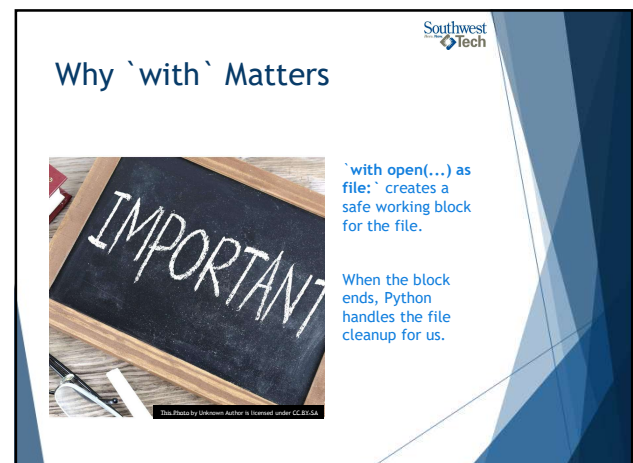
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The 'with' Block Handles the File Safely

- 1 Open file
- 2 Use file inside block
`with open(...) as file:`
- 3 File closes after block

✓ 'with' gives the file a safe working space.

Using the "with" block

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File Modes Are Intent Signals

The file mode tells Python what we intend to do.

- ▶ `"w"` means write
- ▶ `"r"` means read

Wrong mode, wrong behavior.

(Note there are other modes, but we will learn those later)

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File Modes Signal Intent

"w"

write

put data into a file

note.txt

"r"

read

bring saved data back

Choose the mode that matches what the program is trying to do.

File Modes

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Demo #1

"Write A Text File"

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What to Watch For

- ▶ the file name
- ▶ the text being written
- ▶ the file appearing or changing on disk

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write text

```
with open('notes.txt', 'w') as file:
    file.write('Remember this task\n')
    file.write('Review later\n')
```

The program writes text to a file.

notes.txt

Remember this task
Review later

New file created with your text.

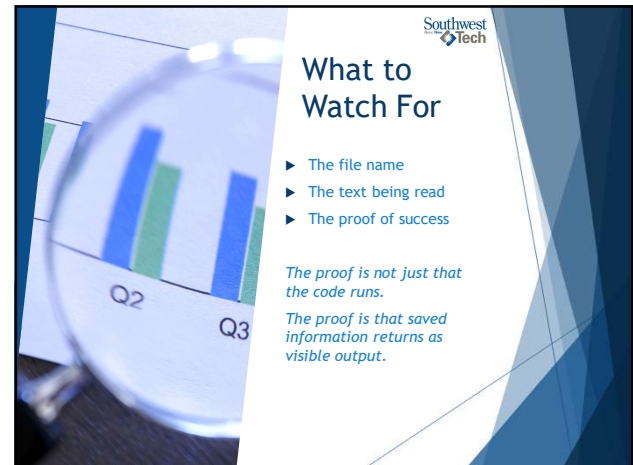
The text you write is saved in a new file.

Demo 1: File Write

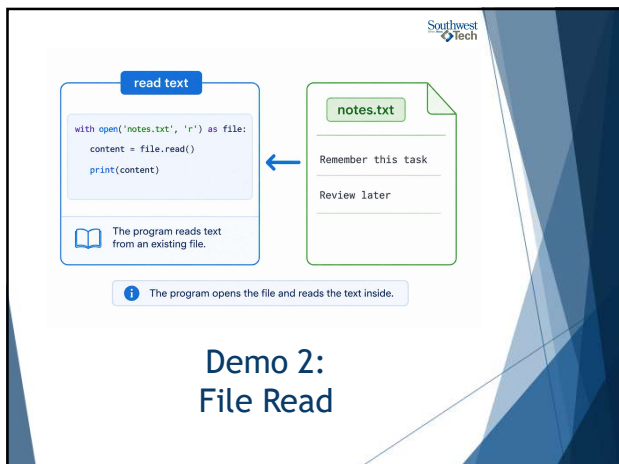
24



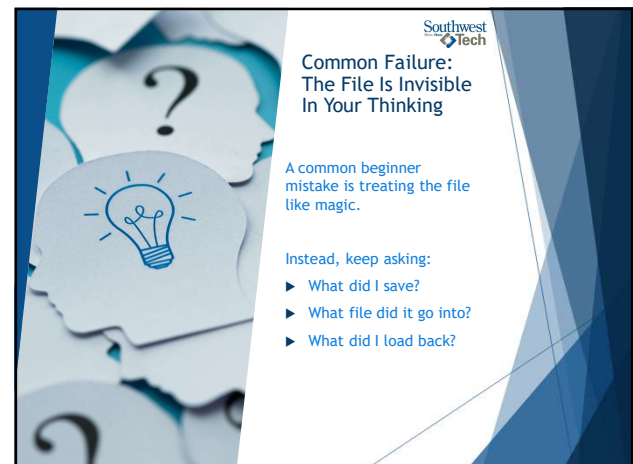
25



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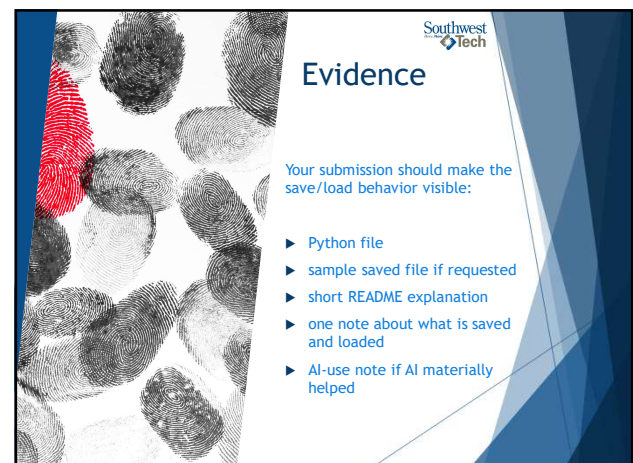
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Assignment 8 Evidence

Include these items to show complete work.

	Python file	☐
	saved data file	☐
	run output	☐
	README explanation	☐
	AI-use note, if used	☐

☒ These items help show your work clearly.

Assignment Evidence

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Closing Success Check

By the end of today,
you should be able to say:

*"My program saved this
information in this file, then
loaded it back and showed it
clearly."*



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Thank you!

Have Fun!

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